

Quick Activity 2: Canteen Group Order

Problem Analysis

Input:

- Number Of Rice Meals
- Number Of Drinks
- Payment

Processes:

1. Multiply the number of Rice Meal by ₱85.
2. Multiply the number of drinks by ₱25.
3. Multiply the subtotal of rice meals by 10% discount.
4. Multiply the number of Rice meal by × ₱5/Packaging fee.
5. Add the two costs to get the final bill.
6. Subtract the final bill from the payment to get the change.

Outputs:

- Final Bill
- Change

Pseudocode:

START					
READ	ricemeals,	drinks,		payment	
ricemealsSubtotal	-->	notebooks	×	₱85	
drinksSubtotal	-->	pen	×	₱25	
discount	-->	mealcost	×	10%	
Packaging	-->	numberofricemeals	×	₱5	
finalbill	-->	ricemealsSubtotal	+	drinkSubtotal	
change	-->	payment	-	finalbill	
DISPLAY	finalbill, change				
End					

Dry-Run test 1

Given:

- 3 rice meals × ₱85
- 4 drinks × ₱25
- Payment = 500
- Discount 10%
- Packaging fee = ₱5

Calculations:

Step	Variable	Calculation	Solution
1	Rice meal	3 x ₱85	₱255
2	Drink	4 x ₱25	₱100
3	Discount	₱255 x 0.10	₱25.50
4	Packaging Fee	3 x ₱5	₱15
5	Final cost	₱255 - ₱25.50 + ₱100 + ₱15	₱344.50
6	change	₱500 - ₱344.50	₱155.50

Final Bill: ₱344.50

Change: ₱155.50

Dry-Run test 2

Given:

- 5 rice meals $\times$ ₱85
- 4 drinks $\times$ ₱25
- Payment = ₱600
- Discount 10%
- Packaging fee = ₱5

Calculations:

Step	Variable	Calculation	Solution
1	Rice meal	$5 \times \text{₱}85$	₱425
2	Drink	$3 \times \text{₱}25$	₱75
3	Discount	$\text{₱}425 \times 0.10$	₱42.50
4	Packaging Fee	$5 \times \text{₱}5$	₱25
5	Final cost	$\text{₱}425 - \text{₱}42.50 + \text{₱}75 + \text{₱}25$	₱482.50
6	Change	$\text{₱}600 - \text{₱}482.50$	₱117

Final Bill: ₱482.50

Change: ₱117.50

Explanation:

Applying the 10% discount to the whole bill would be wrong because the discount is only for the rice meals. The drinks and packaging fees don't have a discount. If we apply the discount to everything, the total discount would be too much, making the final bill incorrect.

